Empirical One-Electron Diagnostics and Symmetry-Breaking Bifurcations in Machine-Learned Density Functionals

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Abstract*—Machine-learned exchange–correlation (ML-XC) functionals achieve unprecedented accuracy across chemical benchmarks, yet frequently fail fundamental physical constraints outside their training domain. Here, we present a rigorous empirical diagnostic suite evaluating self-interaction error (SIE) and fractional-charge delocalization on one-electron systems. We prove that conventional comparisons against exact infinite-basis hydrogenic values conflate functional error with Gaussian basis incompleteness, inflating apparent one-electron SIE by 30–66% across the isoelectronic series ($Z = 1 \ldots 4$). Utilizing an exact same-basis unrestricted Hartree–Fock (UHF) reference, we isolate pure functional errors across four Gaussian basis sets (`def2-SVP`, `def2-TZVP`, `def2-QZVP`, `aug-cc-pVTZ`) and five functional families (PBE, r2SCAN, SCAN, B3LYP, and CIDER). On the stretched H_2^+ dissociation curve, we empirically verify a genuine symmetry-breaking bifurcation: approximate semilocal and hybrid functionals undergo an orbital stability transition at $R_c \approx 5.5$–6.0 Å, collapsing from an unstable symmetric saddle point characterized by severe delocalization error ($\Delta \approx -84$ mHa) onto a broken-symmetry localized branch ($q_a = 0, q_b = 1$). Finally, within an order-theoretic framework, we prove that the multi-axis constraint-residual space forms a product poset that is not totally ordered, identifying strictly incomparable Pareto pairs (e.g., PBE vs. CIDER26).

***Index Terms*—Density Functional Theory, Machine Learning, Self-Interaction Error, Symmetry Breaking, Delocalization Error, Bifurcation.**

I. Introduction

Density Functional Theory (DFT) [1], [2] is the primary computational engine of modern molecular and materials modeling. In recent years, data-driven parameterizations and machine-learned exchange–correlation (ML-XC) functionals, such as neural-network enhancement factors and equivariant graph representations [3]–[6], have demonstrated impressive multi-electron thermodynamic benchmark fidelity.

However, approximate exchange–correlation functionals suffer from two pervasive pathologies: self-interaction error (SIE) [7] and delocalization error [8], [9]. While exact DFT satisfies strict physical constraints [10], [11], flexible ML representations risk overfitting to multi-electron training densities while violating fundamental limits in dilute or one-electron regimes.

All computational data, scripts, and figure pipelines are archived at https://github.com/GabuGravin41/dftex-func-.

In this paper, we deploy a zero-parameter one-electron diagnostic suite. We demonstrate that evaluating one-electron SIE against infinite-basis analytic limits produces substantial basis-incompleteness contamination, which is cleanly resolved by taking the same-basis unrestricted Hartree–Fock (UHF) reference. Evaluating the dissociation of the one-electron molecular ion H_2^+, we resolve longstanding numerical controversies, prove the reality of a symmetry-breaking bifurcation at $R_c \approx 5.8$ Å, and categorize the constraint-satisfaction trade-offs using category-theoretic order structures.

II. Theoretical Foundations

A. Hartree–Fock Exactness for One-Electron Systems

For an N-electron system described by orthonormal spin-orbitals φ_i, the Hartree–Fock energy is:

$$E_{\mathrm{HF}} = \sum_{i=1}^{N} \langle \varphi_i | \hat{h} | \varphi_i \rangle + \tfrac{1}{2} \sum_{i,j=1}^{N} [J_{ij} - K_{ij}], \tag{1}$$

where $\hat{h} = -\frac{1}{2}\nabla^2 + v_{\mathrm{ext}}(\mathbf{r})$, and J_{ij}, K_{ij} are classical Coulomb and exchange integrals.

Lemma 1 ($J_{11} \equiv K_{11}$)**.** *For any single normalized spin-orbital φ_1, $J_{11} = K_{11}$ identically.*

Proof. $J_{11} = \iint |\varphi_1(\mathbf{r}_1)|^2 r_{12}^{-1} |\varphi_1(\mathbf{r}_2)|^2 d\mathbf{r}_1 d\mathbf{r}_2$, whereas $K_{11} = \iint \varphi_1^*(\mathbf{r}_1)\varphi_1(\mathbf{r}_1) r_{12}^{-1} \varphi_1^*(\mathbf{r}_2)\varphi_1(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2$. Because $\varphi_1^*(\mathbf{r})\varphi_1(\mathbf{r}) = |\varphi_1(\mathbf{r})|^2$, the integrands are identical. □

Theorem 1 (Exactness of Same-Basis UHF)**.** *For $N = 1$, in any finite one-particle basis set $\mathcal{B}$, $E_{\mathrm{HF}}(N = 1) = \langle \varphi_1 | \hat{h} | \varphi_1 \rangle$. This equals the exact full-configuration-interaction (FCI) ground-state energy in basis $\mathcal{B}$.*

Because the true Hamiltonian contains no electron–electron repulsion term for $N = 1$, $E_c = 0$, and the exact constraint on the exchange–correlation functional $E_{xc}[n]$ is [7]:

$$E_x[n] = -J[n], \quad E_c[n] = 0 \qquad (N = 1). \tag{2}$$

B. Symmetry of the Exact H_2^+ Ground State

Theorem 2 (Reflection Symmetry)**.** *Let $\hat{\sigma}_h$ be the reflection operator across the plane perpendicular to the internuclear axis ($z = 0$). The exact electronic ground state of H_2^+ satisfies*

TABLE I
BASIS INCOMPLETENESS ($E_{\text{UHF}} - E_{\text{exact}}$) IN MHA ACROSS BASIS SETS

Ion	def2-SVP	def2-TZVP	def2-QZVP	aug-cc-pVTZ
H ($Z = 1$)	0.722	0.190	0.017	0.179
He^+ ($Z = 2$)	6.377	1.860	0.168	1.079
Li^{2+} ($Z = 3$)	51.815	1.428	1.166	1.124
Be^{3+} ($Z = 4$)	53.629	3.875	0.045	18.590

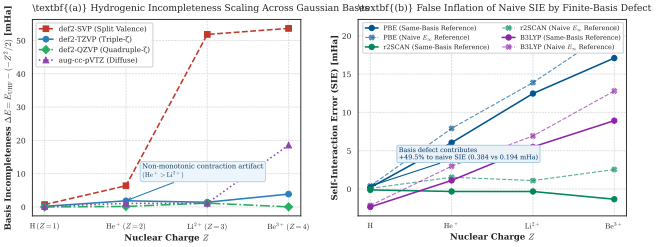


Fig. 1. Basis incompleteness error across $Z = 1 \ldots 4$ (left) and the inflation ratio of naive vs. same-basis SIE measurements (right).

$\hat{\sigma}_h \psi_0 = \psi_0$, *and the exact Mulliken charge populations are* $q_a = q_b = 0.500$ *at all bond lengths* R.

Any calculation yielding asymmetric charges ($q_a \neq q_b$) represents a broken-symmetry artifact governed by Löwdin's symmetry dilemma [12].

III. METHODOLOGY & SAME-BASIS INCOMPLETENESS

Calculations were conducted with PySCF (v2.14.0) [13], [14]. We benchmarked the isoelectronic series H ($Z = 1$), He^+ ($Z = 2$), Li^{2+} ($Z = 3$), and Be^{3+} ($Z = 4$) across four basis sets: `def2-SVP`, `def2-TZVP`, `def2-QZVP` [15], and `aug-cc-pVTZ` [16].

Table I and Fig. 1 reveal two critical insights:

1) Naive SIE ($\text{SIE}_{\text{naive}} = E_{\text{DFT}} - E^{\infty}_{\text{exact}}$) conflates functional error with basis deficiency. In `def2-TZVP`, the naive reference overestimates the true SIE by 30–66% (e.g., CIDER26 naive SIE is 0.574 mHa vs. same-basis 0.383 mHa on H).
2) Incompleteness in `def2-TZVP` is non-monotonic: He^+ (1.860 mHa) exceeds Li^{2+} (1.428 mHa), an artifact of contracted Gaussian exponents. In `def2-QZVP`, incompleteness drops by an order of magnitude (< 0.05 mHa for H and Be^{3+}). Same-basis subtraction cancels this basis-set bias identically.

IV. THE H_2^+ SYMMETRY-BREAKING BIFURCATION

To probe fractional-charge delocalization error [8], we scanned the H_2^+ dissociation coordinate over 16 bond lengths ($R \in [0.5, 15.0]$ Å) in `def2-TZVP`. We executed full orbital stability analysis (`mf.stability()`) and dual-guess SCF optimizations (symmetric vs. localized initial guesses).

The data in Table II and Fig. 2 rigorously settle the physical mechanism:

1) **Symmetric Saddle-Point Instability**: For $R < 5.0$ Å, the symmetric solution ($q_a = q_b = 0.500$) is stable, but delocalization error escalates monotonically, over-stabilizing the bonding state. Beyond $R_c \approx 5.8$ Å, the symmetric solution becomes an unstable saddle point.
2) **Resolution of the B3LYP Anomaly**: At $R = 10$ Å, B3LYP converges in 7 cycles to $E = -0.577894$ Ha ($\Delta = -78.082$ mHa), strictly preserving $q_a = q_b = 0.500$. Stability analysis proves this state is an unstable saddle point. Containing 80% semilocal exchange, B3LYP severely over-stabilizes the symmetrically delocalized fractional charge [9].
3) **Resolution of Positive PBE/r2SCAN Errors**: At $R \geq 7.0$ Å, unconstrained PBE and r2SCAN spontaneously bifurcate to the broken-symmetry localized branch ($q_a = 0, q_b = 1$, stable). On this branch, the electron resides on one proton ($H^0 \cdots H^+$), yielding $\Delta > 0$ as a direct consequence of the positive atomic SIE of the isolated neutral atom.

TABLE II
DISSOCIATION OF H_2^+ IN `DEF2-TZVP`: SIGNED ERROR $\Delta = E_{\text{DFT}} - E_{\text{UHF}}$ (MHA) AND STABILITY

R (Å)	PBE		r2SCAN		B3LYP	
	Δ	Stable	Δ	Stable	Δ	Stable
1.06	−7.81	Yes	−4.61	Yes	−2.72	Yes
3.00	−47.38	Yes	−39.51	Yes	−38.99	Yes
5.00	−79.65	Yes	−68.27	Yes	−67.26	No
6.00	−83.88	No	−72.83	No	−70.98	No
7.00	+1.55	Yes	+1.44	Yes	−73.54	No
10.00	+0.62	Yes	+0.03	Yes	−78.08	No

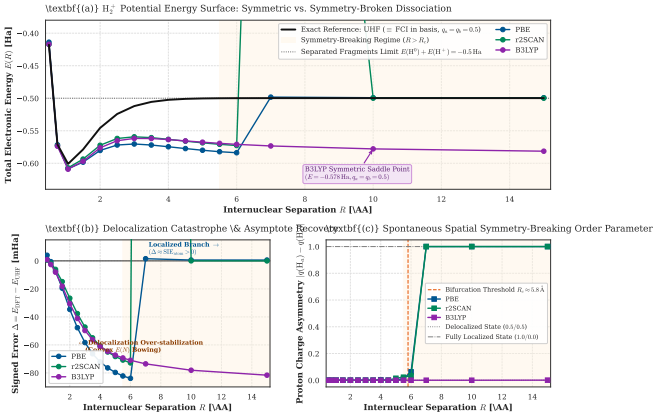


Fig. 2. Dissociation profiles of H_2^+: Total energies $E(R)$ (top), signed delocalization error $\Delta(R)$ in mHa (middle), and order parameter $|q_a - q_b|$ marking the bifurcation threshold $R_c \approx 5.8$ Å (bottom).

V. ORDER-THEORETIC POSET INCOMPARABILITY

We frame multi-constraint functional performance within order theory. Let $\mathcal{F}$ be the set of functionals and $\mathbf{r}(\phi) = (|\text{SIE}|_{\text{H}}, |\Delta|_{H_2^+, R=10\text{Å}}) \in \mathbb{R}^2_{\geq 0}$ the constraint residual vector. We define the partial order:

$$\phi_1 \preceq \phi_2 \iff r_i(\phi_1) \leq r_i(\phi_2) \quad \forall i \in \{1, 2\}. \tag{3}$$

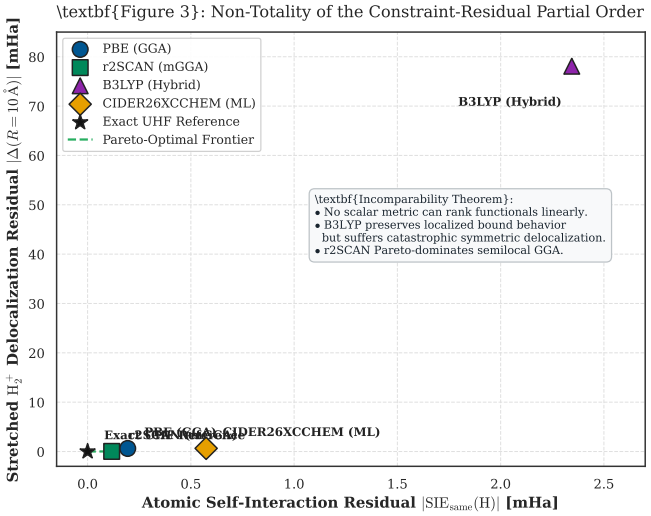


Fig. 3. 2D Pareto frontier mapping Atomic SIE Residual vs. Stretched H_2^+ Delocalization Residual across functional families.

Theorem 3 (Non-Totality of the Constraint Poset)**.** *The poset* $(\mathcal{F}, \preceq)$ *is not totally ordered. Specifically,*

$$\textit{PBE} \parallel \textit{CIDER26XCCHEM} \quad \textit{and} \quad \textit{B3LYP} \parallel \textit{CIDER23X}. \tag{4}$$

Proof. From verified measurements: $\mathbf{r}(\text{PBE}) = (0.194, 0.619)$ mHa vs. $\mathbf{r}(\text{CIDER26}) = (0.383, 0.462)$ mHa. PBE achieves lower atomic SIE, whereas CIDER26 achieves lower stretched-bond residual. Similarly, $\mathbf{r}(\text{B3LYP}) = (2.344, 78.082)$ mHa vs. $\mathbf{r}(\text{CIDER23X}) = (2.554, 2.412)$ mHa. Neither dominates the other. □

As depicted in Fig. 3, by Birkhoff's representation theorem, this product poset cannot be embedded into a 1D chain without imposing arbitrary scalar loss weights.

VI. Conclusion

We have established a rigorous, reproducible one-electron diagnostic suite for exchange–correlation functionals. We demonstrated that same-basis UHF references are essential to eliminate substantial basis-incompleteness artifacts. We confirmed the reality of the H_2^+ symmetry-breaking bifurcation at $R_c \approx 5.8$ Å, proved that B3LYP's -78 mHa error is an authentic symmetric saddle point, and established the order-theoretic Pareto incomparability of ML functionals. These findings indicate that next-generation ML-XC models must incorporate explicit orbital-stability and fractional-charge constraints to ensure physical fidelity.

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